CCNA 200-301 V2.0

27 Wireless Principles

Part V — Wireless, Virtualisation and Operations

EXAM TOPICS

1.5

LECTURER

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THE COURSE

Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	**Wireless, Virtualisation and Operations**	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

LEARNING OUTCOMES

What you will be able to do

- **Select** a band and a channel plan, and explain why 2.4 GHz has only three usable channels.
- **Describe** the RF characteristics that decide whether a link works: signal strength, noise, SNR, and what happens to radio waves on the way.
- **Compare** WEP, WPA, WPA2 and WPA3, and distinguish personal from enterprise authentication.
- **Identify** causes of interference, and tell co-channel interference apart from adjacent-channel interference.

ASSUMED KNOWLEDGE

Before we start

Chapter 3 — wireless is a shared medium with collision *avoidance* rather than detection, and the contrast is the quickest way in. Chapter 9 for how access points attach to the wired network. Chapter 23 for RADIUS, which is what “enterprise” wireless authenticates against.

KEY TERMS

Vocabulary for this chapter

band	channel	non-overlapping	channel bonding	dBm	RSSI	SNR	noise floor	attenuation
SSID	BSSID	WPA2	WPA3	SAE	802.1X	co-channel interference		

Each is defined where it first matters — not here.

ON THE EXAM

Four named sub-topics, and the verb is describe

This is what the blueprint actually asks for

Topic **1.5** is *describe wireless principles*, with four sub-topics named explicitly: **1.5.a** band and channel selection, **1.5.b** RF characteristics, **1.5.c** security protocols, **1.5.d** cause of interference.

There is **no wireless configuration on this blueprint.** v1.1 asked you to build a WLAN in a controller GUI; v2.0 does not. Do not spend time learning WLC menus for this exam — spend it on the four lists in this chapter, because that is exactly what is examined.

Access points do appear elsewhere: **2.2** covers connecting them to a switch port, which is Chapter 9's material, and **1.6** covers the client end, which is Chapter 28's.

DEFINITION

Four names that get confused

- **SSID** — the network's name, as a human sees it. Several APs normally share one.
- **BSSID** — the MAC address of one radio on one AP. This is what a client is actually associated *to*.
- **BSS** — one AP's coverage area. **ESS** — several APs sharing an SSID so a client can roam between them.
- **Roaming** — a client deciding, on its own, to move from one BSSID to another within the same ESS. *The client decides*, not the network. This matters in Chapter 28.

Bands and channels

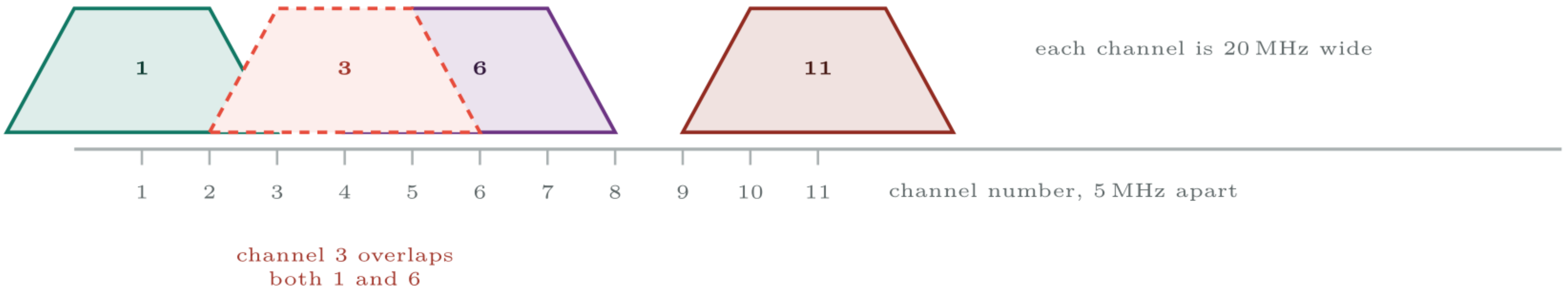
WATCH OUT

Why 2.4 GHz has only three channels

The band is divided into channels numbered 1 to 11 in North America (1 to 13 in much of the rest of the world), spaced 5 MHz apart. But each channel is **20 MHz wide**. A 20 MHz signal spaced 5 MHz from its neighbour overlaps it heavily.

To avoid overlap you need five channel numbers of separation — which in a band of eleven gives you exactly three: **1, 6 and 11**. Any other plan puts two networks partly on top of each other, which is worse than putting them fully on top of each other. That last point is the one the exam tests, and it is explained below.

Bands and channels



The 2.4 GHz band. Channels 1, 6 and 11 are the only trio that does not overlap; anything between them interferes with two networks instead of sharing one.

WATCH OUT

The trade-off, which is examinable

Every doubling of channel width roughly doubles throughput — and **halves the number of non-overlapping channels available**, and raises the noise floor by about 3 dB because the receiver is listening to twice as much spectrum.

So: bonding is sensible at 5 and 6 GHz where there is room, and in a low-density deployment. It is **never** sensible at 2.4 GHz, where a 40 MHz channel consumes most of the band and leaves no clean plan for anyone, including you.

CONCEPT

DFS channels

Part of the 5 GHz band is shared with weather and military radar. Access points using those channels must run **dynamic frequency selection**: listen before transmitting, and vacate the channel within seconds if radar appears.

The practical consequence is a real fault you will meet: a DFS event moves an AP to a different channel and every client on it is briefly disconnected. If a site reports unexplained drops on 5 GHz at irregular intervals — particularly near an airport or a coast — DFS is the first thing to check.

RF characteristics

WORKED EXAMPLE

The two rules that replace a calculator

The 3 dB rule: +3 dB doubles power, -3 dB halves it. **The 10 dB rule:** +10 dB multiplies by ten, -10 dB divides by ten.

So 20 dBm is 10 × 10 × 1,textmW = 100,mW. And 23 dBm is double that again — 200 mW.

Going the other way: a client receiving -70 dBm is receiving one ten-millionth of a milliwatt. Radio receivers are extraordinarily sensitive, which is why small amounts of noise matter so much.

WATCH OUT

Signal strength alone tells you almost nothing

A client at -65 dBm sounds healthy. If the noise floor on that channel is -70 dBm, the SNR is 5 dB and the link is unusable.

A client at -75 dBm sounds marginal. With a noise floor of -95 dBm the SNR is 20 dB and it works perfectly well.

SNR is the number that decides whether a link works. Users and helpdesk tools report bars and RSSI; you should ask for the noise floor before drawing any conclusion. This is the single most useful idea in the chapter and it comes up again in Chapter 28.

CONCEPT

Antennas do not amplify

An antenna with 9 dBi of gain does not create energy. It *shapes* the same energy into a narrower pattern — more in the direction you want, less everywhere else.

- **Omnidirectional** — radiates in a doughnut around the antenna. Ceiling APs in offices. Low gain, wide coverage.
- **Directional** (patch, Yagi, parabolic) — radiates into a cone. Point-to-point links, long corridors, lecture theatres, warehouse aisles. High gain, narrow coverage.

A common failure at scale: replacing omnidirectional antennas with high-gain ones to “fix” coverage, and discovering the new pattern misses the floor directly below the AP entirely.

Security protocols

ON THE EXAM

The two distinctions to get right

This is what the blueprint actually asks for

Personal against Enterprise is about *who* authenticates.

- **Personal (PSK)** — one shared passphrase for everyone. Simple. Nobody is individually identified, and changing it means changing it on every device.
- **Enterprise (802.1X/EAP)** — each user authenticates individually against a RADIUS server (Chapter 23). Revoke one account without touching anyone else. This is what a campus should run.

WPA2 against WPA3 is about *how the key is agreed*. WPA2-Personal's 4-way handshake can be captured and attacked offline, so a weak passphrase falls. WPA3 replaces it with SAE, where an attacker must guess against the live network each time — so offline attack is not possible, and each session has its own key even if the passphrase is later disclosed.

CONCEPT

Open networks and OWE

An open network with no passphrase sends everything in clear. **Opportunistic Wireless Encryption**, part of the WPA3 family and marketed as “Enhanced Open”, encrypts the traffic anyway using an unauthenticated key agreement.

It stops passive eavesdropping on a guest network. It does **not** authenticate the AP, so it does not stop somebody standing up a fake one. Worth deploying on guest SSIDs; not worth mistaking for security.

Causes of interference

WATCH OUT

Counter-intuitive, and a favourite question

Co-channel interference is better than adjacent-channel interference.

Two APs on channel 6 can hear one another, so CSMA/CA works: they defer to each other and share the channel. Throughput halves; nothing is corrupted.

Two APs on channels 6 and 8 cannot decode one another — each sees the other only as raised noise. Neither defers, both transmit, and frames are destroyed and retransmitted. Throughput collapses.

So a bad channel plan is worse than no channel plan. If you take one thing from this section: **overlap, do not straddle.**

CONCEPT

Non-Wi-Fi interference needs a spectrum analyser

A Wi-Fi analyser on a laptop shows you Wi-Fi. A microwave oven does not appear in it, because it transmits no frames — it just raises the noise floor.

So a channel that looks empty in a Wi-Fi scan and performs terribly is the signature of a non-Wi-Fi source. Finding it needs a spectrum analyser, which looks at raw RF energy rather than at decoded frames. Knowing *that the distinction exists* is the examinable part.

Common pitfalls

- **Using any 2.4 GHz channel other than 1, 6 or 11.** Creates ACI for two networks instead of sharing with one.
- **Bonding channels at 2.4 GHz.** Consumes the band.
- **Reading RSSI without the noise floor.** SNR is the number that matters.
- **Turning transmit power up to fix coverage.** In a dense deployment this usually makes things worse, and it does nothing for the client's weaker transmitter on the return path.
- **Confusing SSID with BSSID.** A client roams between BSSIDs while the SSID never changes.
- **Assuming the AP controls roaming.** The client decides.

Common pitfalls (continued)

- **Treating WPA2-Personal as adequate for a campus.** Use Enterprise; a shared passphrase cannot be revoked for one person.
- **Expecting a Wi-Fi scanner to reveal a microwave oven.**

One lecture theatre is unusable when full and perfect when empty.

A 200-seat theatre has four access points and tests clean every morning. During lectures, clients associate but throughput collapses and voice calls fail. The wired network behind the APs shows no errors and no congestion. A survey taken at 08:00 shows every seat at -58 dBm or better.

% A listing nested inside a breakable box cannot itself be split, so % without this tcolorbox forces it onto the page and reports an overfull % vbox. Break the outer box early instead.

One lecture theatre is unusable when full and perfect when empty.

What would you check first — and what do you expect to see?

wifi scan — summary

Survey, taken during a full lecture#

```
BSSID              SSID        Chan  Width  RSSI    Noise   SNR
00:11:22:aa:bb:01  CAMPUS      6     20     -61     -73      12
00:11:22:aa:bb:02  CAMPUS      6     20     -64     -73       9
00:11:22:aa:bb:03  CAMPUS      8     20     -59     -73      14
00:11:22:aa:bb:04  CAMPUS      11    20     -66     -73       7
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Three faults are visible in that one table, and they compound.

First, the noise floor is -73 dBm. A clean environment is around -92. Something is raising it by nearly 20 dB, and the strong signal readings are therefore worthless — every SNR here is below the 20 dB minimum for reliable data.

Second, the channel plan is wrong. Four APs in one room are using channels 6, 6, 8 and 11. The pair on channel 6 is co-channel interference, which is merely wasteful. But **channel 8 overlaps both 6 and 11** — it is adjacent-channel interference against three other radios simultaneously, and that is corruption rather than sharing.

Why it is doing that

Third, the room is full of people. Bodies are largely water, which absorbs 2.4 GHz strongly. This is why it works when empty: the morning survey measured a room with no attenuation in it and no clients contending.

And the lesson

Fix, in order of value.

A survey of somewhere you actually use

Platform note. This topic has no configuration on the blueprint, so this lab is measurement rather than building. Use any Wi-Fi analyser — built-in tools work: `netsh wlan show interfaces` on Windows, the option-click Wi-Fi menu on macOS, `iw dev wlan0 scan` on Linux.

Location. A room you can visit twice, ideally once empty and once busy.

1. Record every visible SSID, its BSSIDs, band, channel and channel width. Draw the channel occupancy of 2.4 GHz as a diagram like the one in this chapter.
2. Identify every case of adjacent-channel interference in what you found. For each, name the channel that should have been used instead.
3. Record RSSI *and* noise for your own connection at five points: beside the AP, across the room, behind a wall, around a corner, and at the far edge of coverage. Calculate SNR at each.
4. Plot RSSI against distance. Confirm the roughly 6 dB loss per doubling of distance, and identify where your measurements depart from it and why.

A survey of somewhere you actually use (continued)

5. Find one place where signal is strong and SNR is poor. Explain what that tells you.
6. Identify the security protocol in use on each SSID you can see. Note any still offering WPA or WEP.
7. Move between two APs on the same SSID while watching the BSSID. Record the RSSI at which your client chose to roam — and note that you did not choose it and the network did not either.
8. **If you can:** repeat the RSSI and noise measurements with the room occupied. Compare, and quantify the absorption.

A survey of somewhere you actually use (continued)

9. **Extension.** Write the one-page recommendation you would give the building's network team, with the evidence for each point.

CHECKPOINT 1 of 6

Why does the 2.4 GHz band have only three non-overlapping channels, and which are they?

Discuss · then advance

ANSWER

Checkpoint 1 of 6

Why does the 2.4 GHz band have only three non-overlapping channels, and which are they?

> Because channels are numbered 5 MHz apart but each is 20 MHz wide, so five channel numbers of separation are needed to avoid overlap. In a band of eleven usable channels that gives exactly three: **1, 6 and 11**.

Which is worse, co-channel or adjacent-channel interference, and why?

Discuss · then advance

ANSWER

Checkpoint 2 of 6

Which is worse, co-channel or adjacent-channel interference, and why?

Adjacent-channel is worse. Two APs on the same channel can hear each other, so CSMA/CA works and they share the airtime — throughput halves but nothing is corrupted. On partly overlapping channels neither can decode the other, so neither defers, both transmit, and frames are destroyed and retransmitted.

A client reports -62 dBm and cannot hold a call. What is the next number you ask for, and what would make this normal?

Discuss · then advance

ANSWER

Checkpoint 3 of 6

A client reports -62 dBm and cannot hold a call. What is the next number you ask for, and what would make this normal?

> The **noise floor**, so you can calculate SNR. A signal of -62 is strong; if the noise floor is around -70 the SNR is only 8 dB, well below the 25 dB voice needs. That would make the report entirely normal.

CHECKPOINT 4 of 6

State the difference between WPA2-Personal and WPA2-Enterprise, and between WPA2 and WPA3.

Discuss · then advance

Checkpoint 4 of 6

State the difference between WPA2-Personal and WPA2-Enterprise, and between WPA2 and WPA3.

Personal shares one passphrase among everyone and identifies nobody; Enterprise authenticates each user individually via 802.1X against a RADIUS server, so one account can be revoked without touching anyone else. WPA2 uses AES-CCMP with a 4-way handshake that can be captured and attacked offline; WPA3 replaces it with SAE, which resists offline attack and adds forward secrecy.

Give two reasons a room might work when empty and fail when full.

Discuss · then advance

Checkpoint 5 of 6

Give two reasons a room might work when empty and fail when full.

> Bodies are largely water, which absorbs 2.4 GHz strongly — so a full room attenuates far more than an empty one. And a full room means many clients contending for the same airtime on a half-duplex shared medium.

CHECKPOINT 6 of 6

What kind of interference is invisible to a Wi-Fi analyser, and what tool finds it?

Discuss · then advance

ANSWER

Checkpoint 6 of 6

What kind of interference is invisible to a Wi-Fi analyser, and what tool finds it?

> Non-Wi-Fi interference — a microwave oven, a video sender, a cordless phone. It transmits no frames, so it never appears in a Wi-Fi scan; it simply raises the noise floor. Finding it needs a **spectrum analyser**, which looks at raw RF energy rather than decoded frames.

What to take away

- Wireless is half duplex, shared, and uses collision *avoidance*. The advertised rate is shared between everyone on the channel.
- SSID is the name, BSSID is the radio's MAC. The *client* decides when to roam.
- 2.4 GHz: three non-overlapping channels, 1, 6 and 11. 5 GHz: around 25, some DFS. 6 GHz: 59, newest clients only.
- Channel bonding doubles throughput, halves the channel count and raises the noise floor. Never at 2.4 GHz.
- +3 dB doubles power, +10 dB multiplies by ten. 0 dBm is 1 mW.

What to take away (continued)

- **SNR decides whether a link works**, not RSSI. Aim for 25 dB for voice, 20 dB minimum for data.
- Radio waves are absorbed, reflected, refracted, diffracted and scattered; free-space loss costs about 6 dB per doubling of distance.
- WEP is broken, WPA deprecated, WPA2 uses AES-CCMP with a 4-way handshake, WPA3 uses SAE and resists offline attack. Personal shares a passphrase; Enterprise authenticates each user via RADIUS.
- Co-channel interference shares politely; adjacent-channel interference corrupts. A bad plan is worse than no plan.

Where we got to

- Wireless is half duplex, shared, and uses collision *avoidance*. The advertised rate is shared between everyone on the channel.
- SSID is the name, BSSID is the radio's MAC. The *client* decides when to roam.
- 2.4 GHz: three non-overlapping channels, 1, 6 and 11. 5 GHz: around 25, some DFS. 6 GHz: 59, newest clients only.
- Channel bonding doubles throughput, halves the channel count and raises the noise floor. Never at 2.4 GHz.

NEXT

Chapter 28 — Troubleshooting Client Connectivity